

Application workflow on patient BraTS2021_00060

1.FIRST PAGE

The application opens up a screen where the user selects a single patient MRI folder for analysis. After a valid folder is chosen, the system confirms the detected format and automatically identifies the four MRI modalities (T1, T1ce, T2, and FLAIR) before processing begins.

Brain Tumor Segmentation and Analysis

Brain Tumor Segmentation and Analysis

Select a patient MRI folder (the 4 modalities), then run.

PATIENT MRI FOLDER

Browse

✓ App assets found.

Inference quality: Fast (overlap 0.25)

Run segmentation and analysis

Show log

Brain Tumor Segmentation and Analysis

Brain Tumor Segmentation and Analysis

Select a patient MRI folder (the 4 modalities), then run.

PATIENT MRI FOLDER

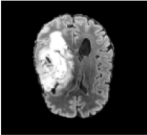
Browse

✓ Valid patient folder

Detected format: BraTS2021

Modality → file

- ✓ T1ce: BraTS2021_00060_t1ce.nii.gz
- ✓ T1: BraTS2021_00060_t1.nii.gz
- ✓ T2: BraTS2021_00060_t2.nii.gz
- ✓ FLAIR: BraTS2021_00060_flair.nii.gz



✓ App assets found.

Inference quality: Fast (overlap 0.25)

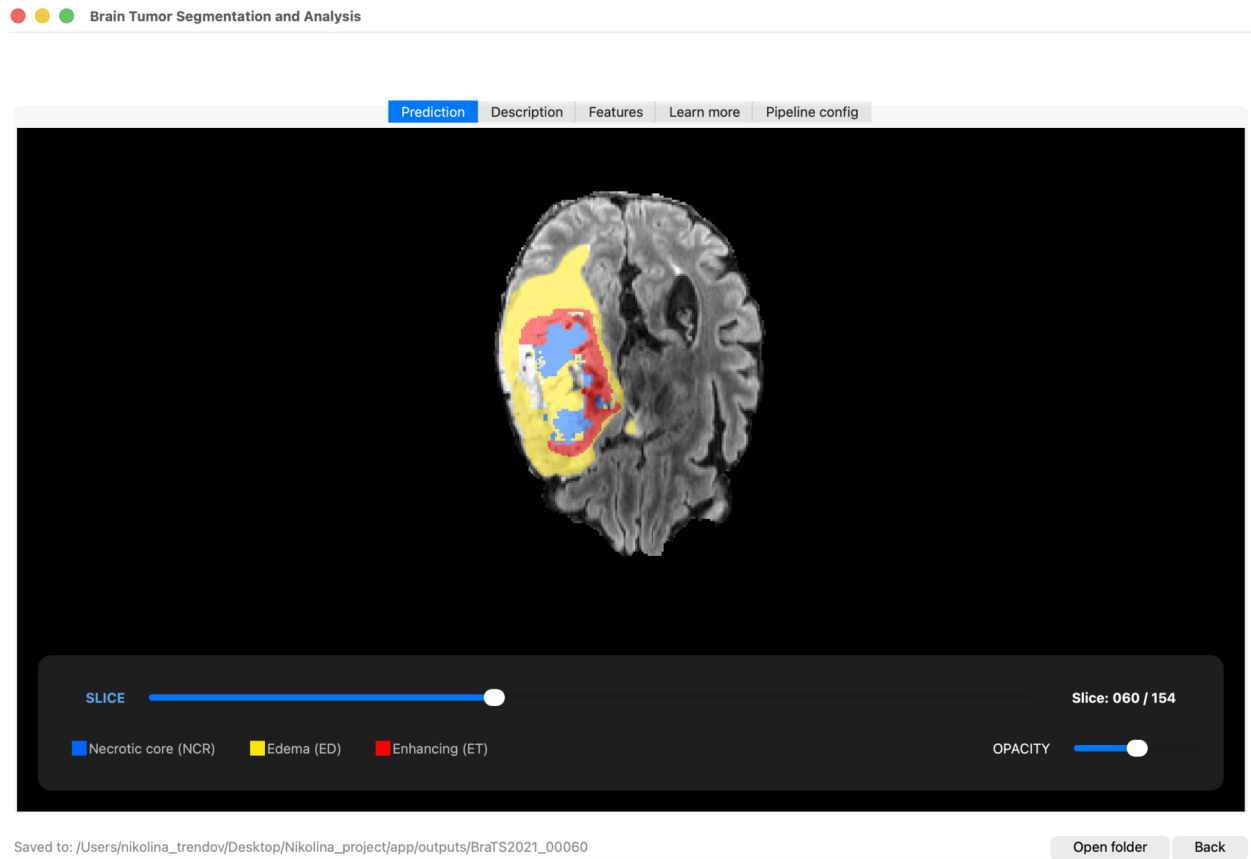
Run segmentation and analysis

Step 5/8: Segmenting (sliding-window inference)

Show log

2. SECTION “PREDICTION”

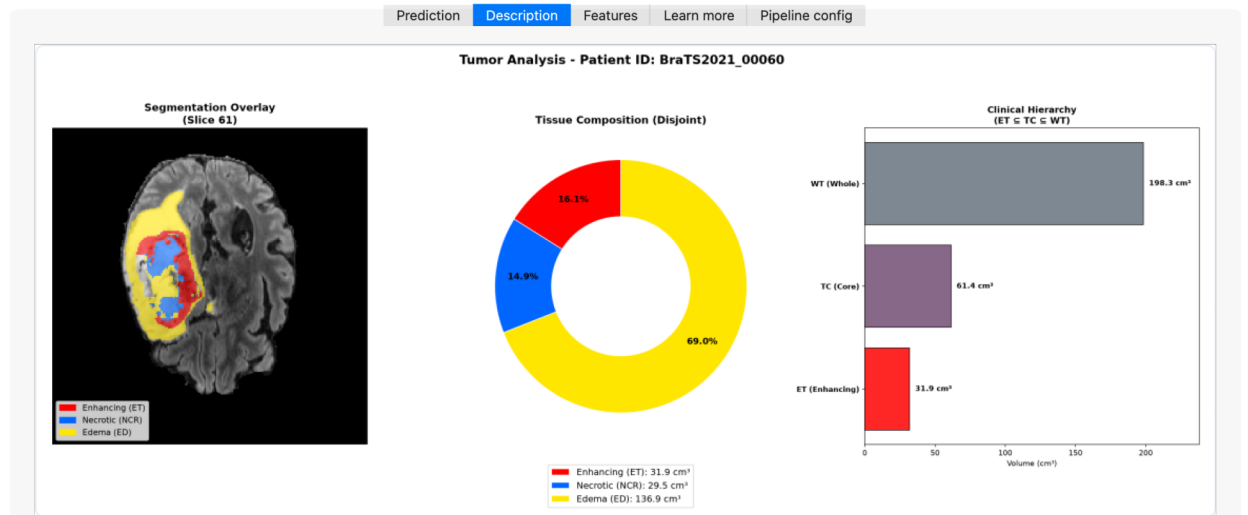
The Prediction tab shows the predicted tumour segmentation overlaid on the MRI. It includes an interactive slice slider for navigating through the volume.



3. SECTION “DESCRIPTION”

The Description tab presents the structured radiological summary. It includes the segmentation overlay, the tissue composition donut chart, the volumetric hierarchy and a table of findings with full provenance tracking.

● ● ● Brain Tumor Segmentation and Analysis



STRUCTURED RADIOLOGICAL DESCRIPTION

Section	Findings	Data Provenance & Logic
Tumor Size	- Tumor core (TC): 61.4 cm³ - Enhancing (ET): 31.9 cm³ - Whole tumor (WT): 198.2 cm³	Source: SwinUNETR Segmentation Directly Measured (Voxel Count / 1000): ET (Enhancing Tumor), TC (Tumor Core), WT (Whole Tumor). Derived via Subtraction: NCR (Necrosis) = TC - ET (61.4 - 31.9 = 29.5 cm³). ED (Edema) = WT - TC (198.2 - 61.4 = 136.9 cm³).
Composition	- Necrosis: extensive (0.481) - Enhancement: minimal (0.161)	Metric Derivation Necrosis Fraction: NCR / TC (29.5 / 61.4 = 0.481). Enhancing Fraction: ET / WT (31.9 / 198.2 = 0.161). Ranked vs cohort tertiles (33rd/66th pct).
Morphology	- Margins: markedly irregular - Sphericity: 0.205	Source: ET Mask (T1ce) Logic: Sphericity measures how closely the shape resembles a sphere (1.0). Classification: 0.205 indicates markedly irregular geometry. Ranked vs cohort tertiles (33rd/66th pct).

Disclaimer: Descriptive only; not a prediction of grade, molecular status or outcome.

4. SECTION “FEATURES”

The Features tab lists the fifteen interpretable radiomic descriptors and volumetric ratios. For each feature, it shows the region, modality, value and unit.

Prediction	Description	Features	Learn more	Pipeline config
Region	Modality	Feature	Value	Unit
Whole Tumor	FLAIR	Voxel Volume	198232.0	mm ³
Whole Tumor	FLAIR	Flatness	0.431	0–1
Whole Tumor	FLAIR	Elongation	0.63	0–1
Tumor Core	T1CE	Skewness	0.549	unitless
Whole Tumor	FLAIR	Surface Volume Ratio	0.235	mm ⁻¹
Enhancing	T1CE	Flatness	0.426	0–1
Enhancing	T1CE	Elongation	0.634	0–1
Tumor Core	T1CE	Surface Volume Ratio	0.381	mm ⁻¹
Whole Tumor	FLAIR	Skewness	-0.176	unitless
Enhancing	T1CE	Sphericity	0.205	0–1
Enhancing	T1CE	Surface Volume Ratio	0.747	mm ⁻¹
Ratio	-	Core Fraction (TC/WT)	0.31	fraction
Ratio	-	Enhancing Fraction (ET/WT)	0.161	fraction
Ratio	-	Necrotic Fraction (NCR/TC)	0.481	fraction
Ratio	-	Enhancing Fraction (ET/TC)	0.519	fraction

5. SECTION “LEARN MORE” (RAG)

The Learn more tab generates, on request, a grounded natural-language explanation of the findings. References are cited to their source and a disclaimer that is restricting the output to descriptive content is added.

● ● ● Brain Tumor Segmentation and Analysis

PredictionDescriptionFeaturesLearn morePipeline config

Research-Based Explanation (RAG)

Generate Research Analysis

Based on the patient's measured values and retrieved references, here is the description of the picture: The MRI image shows a brain tumor with an irregular shape and markedly irregular margins. The margin is difficult to define due to its primary nature. Definition: Margin refers to the boundary between the tumor and surrounding brain tissue. Primary brain tumors rarely have sharp, distinct edges, making it challenging to define the margin on imaging (Source: [Definitions] Margin). The image also shows that the tumor has a sphericity score of 0.205, indicating an irregular shape. Definition: Sphericity is a radiomic feature that measures how closely a tumor's 3D volume approximates a perfect sphere. A lower value like 0.205 indicates more irregular or elongated shapes (Source: [Definitions] Sphericity). Furthermore, the image reveals areas of enhancement within the tumor, which appear brighter on post-contrast T1-weighted MRI. Definition: Enhancement refers to areas that appear brighter on post-contrast T1-weighted MRI due to contrast agent passing through areas of disrupted blood-brain barrier (Source: [Definitions] Enhancement). Disclaimer: Educational explanation of imaging findings only. Not a diagnosis, treatment recommendation or prediction of outcome.

Saved to: /Users/nikolina_trendov/Desktop/Nikolina_project/app/outputs/BraTS2021_00060

Open folderBack

6. SECTION “PIPELINE CONFIG”

The Pipeline config tab documents the exact model, preprocessing, inference, post-processing, radiomics settings used, interpretation and channel order.

Brain Tumor Segmentation and Analysis

[Prediction](#) [Description](#) [Features](#) [Learn more](#) [Pipeline config](#)

System Configuration & Parameters

Model
SwinUNETR(in=4, out=3, feature_size=48) weights: fine_tuning_best_model.pt

Preprocessing
z-score on non-zero voxels, channel-wise; no resampling (BraTS is 1 mm); no skull-strip / N4 (already preprocessed)

Inference
sliding-window roi=(64, 64, 64), overlap=0.25, sigmoid>0.5 -> labels {1:NCR, 2:ED, 4:ET}

Post Processing
largest connected component + fill holes

Radiomics
PyRadiomics Original image, binWidth=25, normalize=False; ET/TC on T1ce, WT on FLAIR

Interpretation
Phase 4 structured description (sizes in cm3, cited composition/morphology bands)

Channel Order
flair, t1ce, t1, t2

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